WEB DEVELOPMENT

ACTIVITY-BASED LEARNING

THREE SELECTED CASE STUDIES

Submitted by

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Selected Case Studies

1. Personal Portfolio Website
2. College Information and Admissions Website
3. Gaming and Esports Community Website

INTRODUCTION

The objective of this Web Development Activity-Based Learning assignment is to apply basic web development concepts by designing and developing webpages using HTML and CSS. The three selected case studies demonstrate how webpage structure, styling, navigation, cards, tables, and responsive layouts can be used to present information clearly.

HTML is used to structure the content of each webpage, while CSS is used to control colors, spacing, typography, layouts, and responsive behavior. This assignment intentionally does not use JavaScript, as requested.

The selected projects are a Personal Portfolio Website, a College Information and Admissions Website, and a Gaming and Esports Community Website. Each project has its own webpage source code and can be opened in a browser by saving the code as index.html.

Objectives

• Build static webpages using semantic HTML elements.
• Apply CSS for visual design and responsive layouts.
• Organize information using sections, cards, navigation links, and tables.
• Practice maintaining separate files for multiple web development projects.

1. Personal Portfolio Website

Save the following code as **index.html** in a separate folder. This project uses HTML and CSS only; JavaScript is not included.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Personal Portfolio | Hemanth</title>
<style>
*{box-sizing:border-box;margin:0;padding:0}body{font-family:Arial,sans-serif;background:#f4f7fb;color:#1e293b;line-
height:1.6}
header{background:#111827;color:white;padding:22px 8%;display:flex;justify-content:space-between;align-items:
center;flex-wrap:wrap;gap:12px}
header h1{font-size:1.5rem;color:#67e8f9}nav a{color:white;text-decoration:none;margin:0 9px;font-size:.95rem}nav
a:hover{color:#67e8f9}
.hero{padding:70px 8%;background:linear-gradient(120deg,#0f172a,#164e63);color:white;display:flex;align-items:
center;justify-content:space-between;gap:30px;flex-wrap:wrap}
.hero h2{font-size:2.5rem;line-height:1.2;margin-bottom:14px}.hero p{max-width:620px;color:#dbeafe}.pill{display:
inline-block;margin-top:20px;background:#67e8f9;color:#083344;padding:10px 18px;border-radius:24px;text-decoration:
none;font-weight:bold}
.avatar{width:150px;height:150px;border-radius:50%;background:#cffafe;color:#155e75;display:grid;place-items:
center;font-size:3rem;font-weight:bold;border:6px solid #a5f3fc}
main{max-width:1100px;margin:auto;padding:35px 20px}.section{background:white;border-radius:14px;padding:25px;
margin:18px 0;box-shadow:0 5px 18px #0f172a0c}
h3{color:#0e7490;margin-bottom:14px;font-size:1.4rem}.grid{display:grid;grid-template-columns:repeat(auto-fit,
minmax(210px,1fr));gap:15px}.card{padding:17px;background:#ecfeff;border-left:4px solid #0891b2;border-radius:8px}.
card h4{margin-bottom:6px}.skills span{display:inline-block;background:#cffafe;color:#155e75;padding:7px 12px;
border-radius:18px;margin:4px}
footer{text-align:center;background:#111827;color:#cbd5e1;padding:20px}@media(max-width:600px){.hero h2{font-size:
2rem}nav a{display:inline-block;margin:5px}}
</style>
</head>
<body>
<header><h1>Hemanth.dev</h1><nav><a href="#about">About</a><a href="#education">Education</a><a
href="#skills">Skills</a><a href="#projects">Projects</a><a href="#contact">Contact</a></nav></header>
<section class="hero"><div><h2>Hello, I'm Hemanth</h2><p>A first-year engineering student interested in web
development, technology, and gaming. I enjoy learning by building simple and useful projects.</p><a class="pill"
href="#projects">View My Work</a></div><div class="avatar">NH</div></section>
<main>
<section class="section" id="about"><h3>About Me</h3><p>I am a curious learner developing my skills in HTML and
CSS. My goal is to create clean, accessible, and responsive websites while improving my programming knowledge.</
p></section>
<section class="section" id="education"><h3>Education</h3><div class="grid"><div class="card"><h4>B.Tech — CSM</
h4><p>First Year</p><p>Focus: Programming and web technologies</p></div><div class="card"><h4>Intermediate</
h4><p>BiPC / relevant previous education</p><p>Building a strong foundation for future learning.</p></div></div></
section>
<section class="section" id="skills"><h3>Technical Skills</h3><div class="skills"><span>HTML5</span><span>CSS3</
span><span>Responsive Design</span><span>Python Basics</span><span>Problem Solving</span></div></section>
<section class="section" id="projects"><h3>Projects</h3><div class="grid"><div class="card"><h4>Portfolio Website</
h4><p>A responsive page to present my profile and skills.</p></div><div class="card"><h4>College Information Page</
h4><p>A simple interface for displaying courses and campus facilities.</p></div><div class="card"><h4>Gaming
Community Page</h4><p>A static page showing games, tournaments, and rankings.</p></div></div></section>
<section class="section"><h3>Hobbies & Interests</h3><p>Gaming, cricket, exploring technology, learning web design,
 and watching movies.</p></section>
<section class="section" id="contact"><h3>Contact</h3><p>Email: hemanth@example.com</p><p>Location: Andhra Pradesh,
 India</p><p><em>Replace the sample email and details with your actual information before submission.</em></p></
section>
</main><footer>© 2026 Hemanth | Personal Portfolio Website</footer>
</body></html>
```

2. College Information and Admissions Website

Save the following code as **index.html** in a separate folder. This project uses HTML and CSS only; JavaScript is not included.

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.0"><title>College
Information Portal</title>
<style>
*{box-sizing:border-box;margin:0;padding:0}body{font-family:Arial,sans-serif;color:#1f2937;line-height:1.6;
background:#f8fafc}
header{background:#12345a;color:#fff;padding:20px 7%;display:flex;justify-content:space-between;align-items:center;
gap:15px;flex-wrap:wrap}
.brand{font-size:1.35rem;font-weight:bold}.brand span{color:#fbbf24}nav a{color:#fff;text-decoration:none;margin:0
8px}nav a:hover{color:#fbbf24}
.hero{padding:70px 7%;background:linear-gradient(120deg,#12345a,#087e8b);color:white}.hero h1{font-size:2.5rem;max-
width:750px;line-height:1.2}.hero p{max-width:700px;margin:16px 0;color:#e0f2fe}.button{display:inline-block;
padding:11px 18px;background:#fbbf24;color:#172554;text-decoration:none;border-radius:6px;font-weight:bold}
main{max-width:1150px;margin:auto;padding:30px 20px}.section{padding:25px 0}.section h2{font-size:1.7rem;color:
#12345a;margin-bottom:15px}.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));gap:18px}.
card{background:white;padding:22px;border-radius:12px;box-shadow:0 4px 15px #0f172a12;border-top:4px solid
#0e7490}.card h3{margin-bottom:8px;color:#0e7490}.highlight{background:#e0f2fe;padding:22px;border-radius:12px}.
steps{padding-left:22px}.steps li{margin:8px 0}
table{width:100%;border-collapse:collapse;background:white}th,td{padding:12px;text-align:left;border:1px solid
#cbd5e1}th{background:#12345a;color:white}
.contact{background:#eaf4f4;padding:22px;border-radius:12px}footer{text-align:center;background:#102a43;color:
#e2e8f0;padding:20px}
@media(max-width:600px){.hero h1{font-size:2rem}nav a{display:inline-block;margin:5px}}
</style></head>
<body>
<header><div class="brand">YSR <span>Engineering College</span></div><nav><a href="#about">About</a><a
href="#courses">Courses</a><a href="#facilities">Facilities</a><a href="#admissions">Admissions</a><a
href="#contact">Contact</a></nav></header>
<section class="hero"><h1>Learn Today. Lead Tomorrow.</h1><p>Welcome to our college information portal. Explore
academic programs, campus facilities, student activities, and admission information in one place.</p><a
class="button" href="#admissions">Admission Information</a></section>
<main>
<section class="section" id="about"><h2>About Our Institution</h2><div class="highlight"><p>Our institution aims
to provide quality education, practical learning opportunities, and a supportive environment for students to
develop technical and professional skills. This is a sample academic website created for the Web Development ABL
assignment.</p></div></section>
<section class="section" id="courses"><h2>Departments & Courses</h2><div class="grid"><article
class="card"><h3>Computer Science</h3><p>Programming, software development, databases, and computing fundamentals.
</p></article><article class="card"><h3>Information Technology</h3><p>Networks, web technologies, information
systems, and application development.</p></article><article class="card"><h3>Electronics</h3><p>Electronic
circuits, communication systems, and embedded technologies.</p></article><article class="card"><h3>Mechanical
Engineering</h3><p>Design, manufacturing, materials, and mechanical systems.</p></article></div></section>
<section class="section" id="facilities"><h2>Campus Facilities</h2><div class="grid"><article
class="card"><h3>Library</h3><p>Learning resources, reference books, and quiet study spaces.</p></article><article
class="card"><h3>Computer Labs</h3><p>Practical sessions with computers and development tools.</p></
article><article class="card"><h3>Sports</h3><p>Activities that support teamwork, fitness, and student life.</p></
article><article class="card"><h3>Student Support</h3><p>Academic guidance, mentoring, and career preparation.</
p></article></div></section>
<section class="section" id="admissions"><h2>Admission Process</h2><ol class="steps"><li>Review the eligibility
requirements for the selected course.</li><li>Complete the relevant application or counselling process.</
li><li>Submit required certificates and supporting documents.</li><li>Confirm admission after document
verification and fee payment.</li></ol><h3 style="margin:20px 0 10px;color:#12345a">Sample Academic Programs</
h3><table><tr><th>Program</th><th>Duration</th><th>Study Area</th></tr><tr><td>B.Tech</td><td>4 Years</
td><td>Engineering and Technology</td></tr><tr><td>M.Tech</td><td>2 Years</td><td>Advanced Engineering</td></
tr><tr><td>Diploma</td><td>3 Years</td><td>Technical Education</td></tr></table><p style="margin-top:
10px"><em>Course details are examples; verify actual programs with the college.</em></p></section>
<section class="section" id="contact"><h2>Contact Information</h2><div class="contact"><p><strong>Admissions
Office</strong></p><p>Email: admissions@example.edu</p><p>Phone: 00000 00000</p><p>Office Hours: Monday–Saturday,
9:00 AM–4:00 PM</p><p><em>Replace sample contact details with verified official information.</em></p></div></
section>
</main><footer>College Information & Admissions Portal | ABL Project</footer>
</body></html>
```

3. Gaming and Esports Community Website

Save the following code as **index.html** in a separate folder. This project uses HTML and CSS only; JavaScript is not included.

```
<!DOCTYPE html>
<html lang="en">
<head><meta charset="UTF-8"><meta name="viewport" content="width=device-width, initial-scale=1.0"><title>Gaming
Arena</title>
<style>
*{box-sizing:border-box;margin:0;padding:0}body{font-family:Arial,sans-serif;background:#0b1020;color:#e5e7eb;line-
height:1.6}
header{padding:20px 7%;background:#111827;display:flex;justify-content:space-between;align-items:center;gap:12px;
flex-wrap:wrap;border-bottom:1px solid #293449}.logo{font-weight:900;font-size:1.5rem;color:#a3e635}nav a{color:
#e5e7eb;text-decoration:none;margin:0 9px}nav a:hover{color:#a3e635}
.hero{padding:75px 7%;background:radial-gradient(circle at 80% 30%,#334155 0,#111827 40%,#0b1020 80%)}.tag{color:
#a3e635;text-transform:uppercase;letter-spacing:3px;font-weight:bold}.hero h1{font-size:3rem;line-height:1.1;
margin:12px 0}.hero p{max-width:650px;color:#cbd5e1}.cta{display:inline-block;background:#a3e635;color:#172033;
text-decoration:none;padding:11px 18px;border-radius:7px;font-weight:bold;margin-top:20px}
main{max-width:1150px;margin:auto;padding:25px 20px}.section{padding:25px 0}.section h2{font-size:1.7rem;margin-
bottom:16px}.grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(220px,1fr));gap:17px}.card{background:
#151e31;border:1px solid #293449;border-radius:12px;padding:20px;transition:transform .2s,border-color .2s}.card:
hover{transform:translateY(-4px);border-color:#a3e635}.emoji{font-size:2rem}.card h3{color:#a3e635;margin:8px 0}.
muted{color:#aab6c8}.badge{display:inline-block;background:#293449;padding:4px 9px;border-radius:20px;font-size:.
8rem;margin-top:10px}
table{width:100%;border-collapse:collapse;background:#151e31;border-radius:10px;overflow:hidden}th,td{padding:13px;
text-align:left;border-bottom:1px solid #293449}th{background:#263449;color:#a3e635}
.news{border-left:4px solid #a3e635;padding:12px 16px;background:#151e31;margin:10px 0;border-radius:5px}footer{
text-align:center;background:#111827;color:#9ca3af;padding:22px;margin-top:20px}
@media(max-width:600px){.hero h1{font-size:2.25rem}nav a{display:inline-block;margin:5px}}
</style></head>
<body>
<header><div class="logo">■ GAMING ARENA</div><nav><a href="#games">Games</a><a href="#tournaments">Tournaments</
a><a href="#leaderboard">Leaderboard</a><a href="#news">News</a></nav></header>
<section class="hero"><span class="tag">Play • Compete • Connect</span><h1>Your Next Match<br>Starts Here.</
h1><p>A community hub for mobile and PC gamers. Discover popular games, follow tournament schedules, and check out
the community leaderboard.</p><a class="cta" href="#tournaments">Explore Tournaments</a></section>
<main>
<section class="section" id="games"><h2>Featured Games</h2><div class="grid"><article class="card"><div
class="emoji">■■</div><h3>Battle Royale</h3><p class="muted">Survival, teamwork, and fast-paced matches.</p><span
class="badge">Mobile / PC</span></article><article class="card"><div class="emoji">■■</div><h3>Racing Legends</
h3><p class="muted">Compete for the fastest lap and best time.</p><span class="badge">Racing</span></
article><article class="card"><div class="emoji">■</div><h3>Football Champions</h3><p class="muted">Build a team
and compete in football matches.</p><span class="badge">Sports</span></article><article class="card"><div
class="emoji">■</div><h3>Adventure World</h3><p class="muted">Explore new worlds and complete challenges.</p><span
class="badge">Adventure</span></article></div></section>
<section class="section" id="tournaments"><h2>Upcoming Community Tournaments</h2><div class="grid"><article
class="card"><h3>Weekend Showdown</h3><p class="muted">Format: Squad • Entry: Free</p><p>Date: To be announced</
p></article><article class="card"><h3>Racing Time Trial</h3><p class="muted">Format: Individual</p><p>Date: To be
announced</p></article><article class="card"><h3>Football Cup</h3><p class="muted">Format: Knockout</p><p>Date: To
be announced</p></article></div></section>
<section class="section" id="leaderboard"><h2>Community Leaderboard</h2><table><tr><th>Rank</th><th>Player</
th><th>Points</th><th>Team</th></tr><tr><td>1</td><td>FireStorm</td><td>980</td><td>Team Nova</td></tr><tr><td>2</
td><td>PixelAce</td><td>910</td><td>Red Wolves</td></tr><tr><td>3</td><td>ShadowPlay</td><td>875</td><td>Team
Nova</td></tr><tr><td>4</td><td>RapidX</td><td>820</td><td>Solo</td></tr></table><p class="muted" style="margin-
top:8px">Sample leaderboard data for demonstration.</p></section>
<section class="section" id="news"><h2>Gaming News</h2><div class="news"><strong>Community Night</strong><p
class="muted">Join fellow players for a friendly game night. Details to be announced.</p></div><div
class="news"><strong>New Player Spotlight</strong><p class="muted">Meet community members, share strategies, and
celebrate teamwork.</p></div></section>
</main><footer>Gaming Arena | A static HTML and CSS community website</footer>
</body></html>
```